# TreeLoom: Learning Local Ancestry from Complete Genealogies

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## Abstract

Local ancestry inference (LAI) reconstructs the ancestral origins of segments in admixed genomes. Genealogies reveal relationships to reference haplotypes, but recombination changes those relationships along the chromosome. A model must combine evidence within each genealogy with genomic continuity while distinguishing persistent sampled chromosomes from inferred ancestors. We introduce TreeLoom, which processes complete local trees with a shared graph neural network and follows sampled haplotypes across trees. Haplotype memory returns genomic context to the tree encoder, while a final recurrent network combines the resulting local representations for ancestry prediction. On eight European development simulations, TreeLoom attains $97.08 \pm 0.57\%$ modern accuracy weighted by genomic span, compared with $94.45 \pm 1.15\%$ for the evaluated ARGMix release and $93.78 \pm 0.84\%$ for a derived four source AncestralPaths endpoint. These pipeline comparisons use common observations and scoring support, with material differences in inputs and training. Matched analysis finds only a small accuracy benefit from memory and lower boundary F1. The results support learning from complete genealogies in this setting and highlight ancestry tract recovery as a distinct requirement beyond accurate genomic assignment.


## 1 Introduction

A chromosome inherited from an admixed population contains segments with different ancestral origins. Local ancestry inference (LAI) assigns an origin to each segment and estimates where those origins change. The resulting ancestry tracts help describe population mixture, the movement of genetic material and differences among genomes. Ancient individuals add a temporal dimension: their chromosomes can reveal ancestry before, during and after major demographic changes.

These assignments depend on how query genomes relate to reference genomes. A genealogy makes those relationships explicit. At a particular position, its branches connect sampled haplotypes through common ancestors. A query may be closely related to one reference in one region and to another reference elsewhere. An ancestral recombination graph (ARG) records this changing history across the chromosome through its local trees. Information about ancestry is therefore distributed both within each tree and along the chromosome.

Methods such as AncestralPaths, ARGMix and ARGformer already exploit genealogical information for ancestry analysis (Irving-Pease et al. 2024; Shanks et al. 2026; Bonet et al. 2026). The question we study is how to organize that information when learning from complete local trees. Evidence outside a query's immediate branch can help interpret its position in a tree. Evidence from neighboring regions may also help, because a change in genealogy does not necessarily imply a change in ancestry.

Continuity has a clear biological anchor at sampled haplotypes. Each haplotype represents the same chromosome copy throughout the sequence. Inferred internal ancestors have less certain correspondence between neighboring trees. A shared descendant group or a reused node identifier does not establish that two inferred nodes represent the same biological individual. A model can therefore use structural similarities between trees while keeping genomic history attached to sampled chromosomes.

TreeLoom follows this design. An upward traversal summarizes descendants, and a downward traversal supplies context from outside each branch. A second application of the same graph neural network (GNN) incorporates local structural features and history carried by sampled haplotypes. A genomic recurrent network then predicts ancestry from the ordered local representations. These are established neural operations arranged around the structure of the LAI problem.

We compare the selected model with completed AncestralPaths and ARGMix pipelines on common European observations and scoring support. We report performance by cohort and examine both correctly assigned genomic span and ancestry boundaries. A matched comparison of sample memory is retained to distinguish the performance of the full fitted model from the much smaller effect of that component.

## 2 Related Work

AncestralPaths predicts genealogical ancestry routes, some of which share a broad source population (Irving-Pease et al. 2024). ARGMix combines graph representations with genomic context (Shanks et al. 2026). ARGformer learns genealogical path embeddings for downstream ancestry analysis (Bonet et al. 2026). FLARE instead estimates local ancestry directly from haplotypes (Browning, Waples, and Browning 2023). Their representations, supervision and prediction units differ, which matters when placing their outputs on a

common endpoint.

Message passing through a tree and recurrence along a sequence provide established ways to combine relational and ordered information. Gated recurrent units (GRUs) supply the sequential operation used here (Cho et al. 2014). Graph Set Transformer alternates graph propagation with context from a collection of graphs (Escrig Molina, Chen, and Probst 2026), while temporal graph state space models retain information on changing graphs (Li et al. 2024). Local genealogies have the additional constraints of genomic order, unequal interval lengths and persistent sampled haplotypes. TreeLoom uses those properties to determine where history is stored and how it enters the local encoder.

# 3 TreeLoom

## 3.1 Observed Evidence and Prediction Task

Let $T_t = (V_t, E_t)$ be the inferred local tree covering interval $I_t$ of physical length $w_t$. A sampled haplotype $q$ appears at node $v_t(q)$ in every tree. The model predicts $p_{tq}$ over four source ancestries: Anatolian farmer (Ana), Caucasus hunter-gatherer (CHG), western hunter-gatherer (WHG) and eastern hunter-gatherer (EHG).

Inputs comprise phased observed variants, inferred topology and node ages, sampling ages, and the known source labels of reference haplotypes. Query labels are masked. Simulated ancestry provides training targets and evaluation truth, but does not define the input features, inferred trees, structural groups or prediction intervals.

A multilayer perceptron (MLP) encodes called alleles relative to observed donor and panel allele frequencies. Mean and maximum pooling summarize this evidence within each tree interval for each haplotype. Reference labels and sampling ages supply further information at sampled nodes. Inferred ages and indicators for sampled nodes and roots describe all nodes, including internal ancestors. Their encoded features initialize a state $b_v$. Learned gates use branch geometry and agreement between observed allele partitions and inferred branches to scale messages.

Figure 1 summarizes the model. The encoder makes two rounds through each complete inferred tree, each comprising an upward traversal and a downward traversal. These are four full traversals, with the same node operator reused at every tree depth rather than a fixed stack of four message passing layers. Sample identities align their outputs across intervals, and each sampled haplotype has a separate genomic sequence.

## 3.2 Information within a Complete Genealogy

For a parent $v$ and child $u$, the upward traversal computes

$$m_{u\to v} = \frac{g_u^{\uparrow} \odot W_{\uparrow} h_u}{\max(1, d_v)}, \tag{1}$$

$$s_v = \sum_{u \in \mathrm{child}(v)} m_{u\to v}, \tag{2}$$

$$h_v = \mathrm{LN}(b_v + \tanh(A_{\uparrow}[b_v, s_v] + a_{\uparrow})). \tag{3}$$

Here $d_v$ is the number of children, $g_u^{\uparrow}$ is a gate from observed features, $\odot$ denotes elementwise multiplication, and brackets denote concatenation. The update combines descendant messages with the node's own features through a residual connection and layer normalization (LN). Dividing by the number of children controls the magnitude of the sum.

The downward traversal brings information from outside the child's branch. Its parent context is

$$c_{v\setminus u} = o_v + b_v + s_v - m_{u\to v}. \tag{4}$$

A gated learned transformation maps this context to the outside state $o_u$, starting from zero at the root. Subtracting the child's upward message prevents that message from returning as outside evidence. Thus $h_v$ summarizes descendants and $o_v$ summarizes the rest of the tree.

## 3.3 Local Structure and Haplotype History

Structural features compare trees without identifying their internal ancestors. Let $C_t(v)$ be the set of sampled haplotypes descended from node $v$. A proper clade contains at least two sampled haplotypes and excludes the complete panel. If a unary chain has the same descendant set, its deepest node is the representative.

A clade is gained when it is absent from the preceding tree and lost when it is absent from the following tree. Two learned residuals use these flags and the persistence of clades in consecutive trees to transform the current node state. They use the current tree's features, with neighboring trees determining where the transformations apply. Their equations are given in the supplement.

Sample memory supplies a different kind of context (Figure 2b). A fixed age cut partitions each tree into groups of haplotypes sharing an ancestor at the cut. A sample older than the cut retains its own sampled node. For group $k$ in tree $t$, a projection of the representative's summed upward and outside states gives $f_{tk}$. Write $k_t(q)$ for haplotype $q$'s group.

A forward exponential average follows the same haplotype through successive trees,

$$z^{\rightarrow}_{tq} = \rho_t z^{\rightarrow}_{t-1,q} + (1-\rho_t) f_{t,k_t(q)}, \qquad \rho_t = \exp(-w_t/\lambda). \tag{5}$$

The initial state is its current context. A reverse scan applies the same rule in the other genomic direction. The learned positive length scale $\lambda$ controls retention in physical distance, so a longer interval contributes more of its current context.

The two directions are averaged. Within each current group, the mean haplotype history is compared with the current group context. A learned projection of that difference is added at the group representative. The second complete graph traversal propagates this contribution together with the structural residuals, using the first traversal's weights and observed gates. History stays attached to sampled chromosomes even when their groups change.

## 3.4 Genomic Context and Ancestry Probabilities

After the second traversal, a projection of the upward and outside states gives a local leaf embedding $e_{tq} \in \mathbb{R}^d$. Stacking leaf embeddings in tree order while preserving leaf identity produces one sequence per sampled chromosome. A one layer bidirectional GRU (BiGRU) processes those embeddings and interval geometry,

$$r_{1:T,q} = \mathrm{BiGRU}\left([e_{tq}, \mathrm{geometry}(I_t)]_{t=1}^{T}\right). \tag{6}$$

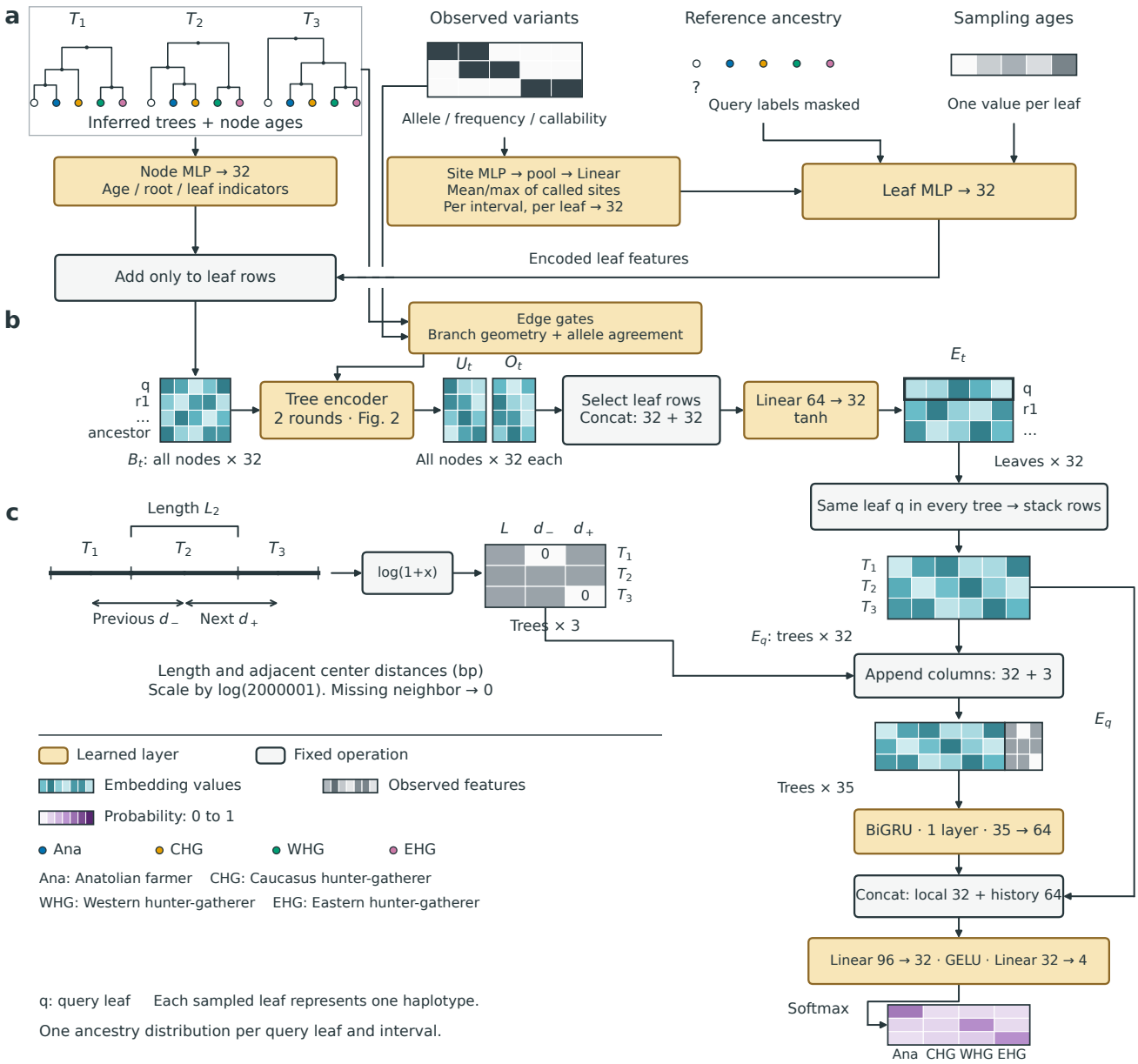



Figure 1: TreeLoom predicts local ancestry for each query leaf and interval. Each sampled leaf represents one haplotype. **a,** Node features initialize all nodes, including ancestors. Observed site features are encoded and pooled over called sites within each interval separately for each leaf. Reference labels and sampling ages join this pooled evidence, and the resulting features are added only to leaf rows. Query ancestry labels are masked. Branch geometry and observed allele–branch agreement supply learned message gates. **b,** The two rounds of the tree encoder are expanded in Figure 2. Its output has upward and outside embeddings for every node. Selecting leaf rows, concatenating the two states and projecting gives a local leaf embedding of width 32. **c,** Rows for the same query leaf are stacked in tree order. Interval length and distances to adjacent interval centers add three columns before the BiGRU. Its output joins the original local embedding for classification. Four ancestry classes illustrate the European condition. All matrix values and probabilities are schematic, not fitted results.

The geometry describes interval length and distances between neighboring interval centers. Each GRU direction has width $h$. The classifier joins the local embedding to the genomic representation,

$$p_{tq} = \mathrm{softmax}\left(\mathrm{MLP}([e_{tq}, r_{tq}])\right). \tag{7}$$

This final recurrence uses genomic context to classify local embeddings. Sample memory instead brings history into the tree encoder before those embeddings are formed.

All sampled haplotypes remain present until query selection at readout. Training minimizes cross entropy weighted

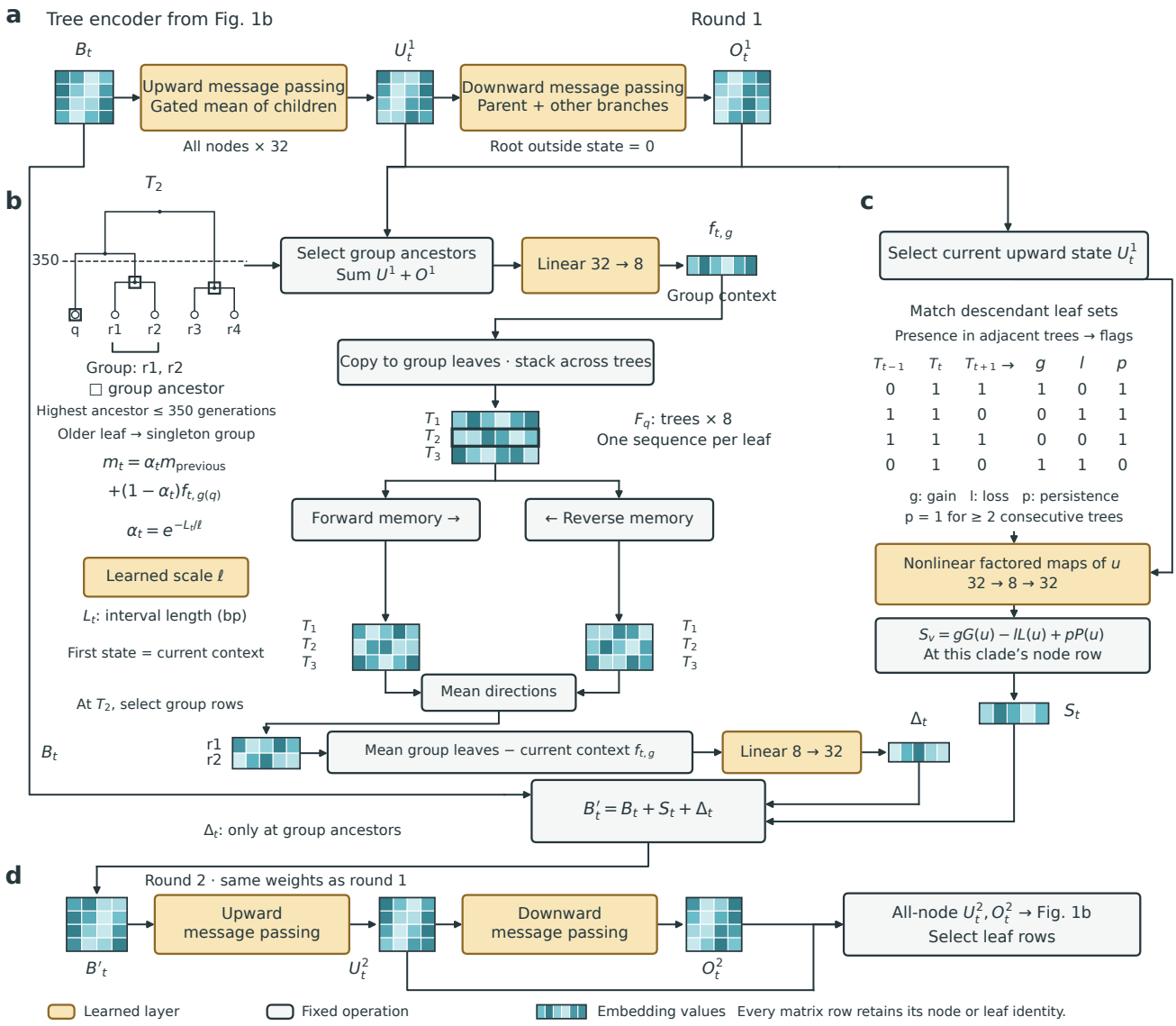



Figure 2: Expansion of the tree encoder in Figure 1b. **a,** A complete upward traversal produces $U^1$, followed by a downward traversal producing $O^1$. Rows retain node identities. Upward state $U$ corresponds to $h$ in the equations. **b,** Each leaf joins the group represented by its highest ancestor no older than 350 generations. Older samples remain singleton groups. A projection of the representative's summed states gives the group context $f$, copied to its leaves. Separate histories follow each leaf across trees, even when its group changes. Forward and reverse exponential memories are averaged per leaf and then over the current group's leaves. Subtracting the current context and projecting gives a correction $\Delta$, added only at group representatives. The $T_2$ example is repeated at every interval. Here $m, \alpha, L, \ell$ correspond to $z, \rho, w, \lambda$ in the text. **c,** Exact descendant sets determine gain, loss and persistence flags for proper clades. These flags may overlap. Learned factored maps $G, L, P$ transform the current upward state and place correction $S$ at its clade representative. Their equations appear in the supplement. **d,** Both corrections enter the original base before the second complete upward and downward round, which reuses the first round's weights. Both all-node outputs return to leaf selection in Figure 1b.

by physical span, with each simulated chromosome as one optimization example. The prediction is constant within an inferred tree interval, so ancestry calls can change only at interval boundaries. Model dimensions and optimization details appear in the supplement.

# 4 Evaluation Design

## 4.1 European Simulations and Scoring

We use 32 training simulations and eight validation simulations, each 2 Mb long, from the released AncestralPaths demographic formulation associated with Irving-Pease et al.

(2024). The panel has 476 diploid individuals, or 952 phased haplotypes, comprising 266 references and 686 queries. Queries include modern individuals, Bronze Age individuals, Bronze Age Anatolians (BAA), Neolithic farmers (Neo) and Yamnaya (Yam). Simulations are separated before tree intervals, rows or sequence chunks are constructed.

Relate infers genealogies from the observed variants and sampling ages (Speidel et al. 2019). The completed European pipelines use this common observed panel and reconstruction in their required representations. The supplement gives the demographic assumptions and inference settings. The Bronze Age roster is only an approximate correspondence to the ARGMix preprint's Post-Neolithic category.

The primary endpoint is accuracy weighted by physical span for 182 modern query haplotypes. We also report ancient cohorts and all queries. Predictions are scored on fixed 100 bp bins clipped to inferred coverage. If $m_{iqc}$ is the physical mass of ancestry $c$ for query $q$ in bin $i$, accuracy is

$$A = \frac{\sum_{iqc} m_{iqc}\, \mathbf{1}\{c = \arg\max_k p_{iqk}\}}{\sum_{iqc} m_{iqc}}. \tag{8}$$

Each simulation has equal weight in the reported mean. Sample standard deviation (SD) across the eight simulations is expressed in percentage points. It describes variation among simulations for one fitted model, not uncertainty across optimizer seeds.

Boundary precision, recall and F1 measure recovery of ancestry transitions. Boundaries are matched one to one within 1 kb by position, without requiring the correct pair of source labels. Negative log likelihood (NLL) uses the same physical masses and the original probabilities, with no flooring or recalibration.

These eight validation simulations have been reused for model development and checkpoint selection. Each condition has one optimizer seed, and independent test evaluation is unfinished. Results therefore describe development performance. The simulation, rather than its dependent haplotypes or genomic bins, is the unit of replication.

### 4.2 What the Comparator Results Measure

AncestralPaths is evaluated using its released dense classifier with six path outputs. We sum path probabilities into four source probabilities before choosing a source,

$$(p_{\mathrm{Ana}}, p_{\mathrm{CHG}}, p_{\mathrm{WHG}}, p_{\mathrm{EHG}}) = (p_1 + p_5, p_2 + p_6, p_3, p_4). \tag{9}$$

This derived task differs from its native six path task. The publication describes a convolutional model, while the release uses a dense model. The five epoch schedule is completed and the final epoch is evaluated. Its replacement sampler leaves 9,126 requested BAA path six rows unfilled, with an unmeasured effect on accuracy.

ARGMix uses the current author release with 15 references and no ancestor inputs. The preprint describes 25 references and three inferred ancestors, a configuration not supported by the released loaders. We adapted reference slots and metadata for the common sample panel while retaining the released model, training augmentation and stopping rule. Training completes 11 epochs, and the native validation criterion selects epoch nine.

| Pipeline | Accuracy ↑ | Boundary F1 ↑ |
|---|---|---|
| AncestralPaths | $93.78 \pm 0.84$ | $7.03 \pm 0.78$ |
| ARGMix, current release | $94.45 \pm 1.15$ | $11.74 \pm 1.73$ |
| TreeLoom | $\mathbf{97.08 \pm 0.57}$ | $\mathbf{18.38 \pm 1.31}$ |

Table 1: Modern European development performance, mean±sample SD in percent across eight 2 Mb simulations and 182 query haplotypes. AncestralPaths probabilities are collapsed from six paths to four source classes. ARGMix uses the current author release. Boundary matching uses a 1 kb positional tolerance. Bold marks the highest mean. Inputs, training and selection differ between pipelines as detailed in the text.

TreeLoom also receives features from observed single nucleotide polymorphisms (SNPs). Reference selection, representation, supervision unit, training diversity, cumulative optimization and checkpoint selection differ among the fitted pipelines. A common scorer makes their outputs comparable on this endpoint, but does not equalize those conditions. These runs are not exact publication reproductions, and their ranking cannot establish an overall ranking of LAI methods.

The component comparison uses TreeLoom-NoMem, which removes sample memory while retaining both complete tree traversals, structural residuals and the final genomic GRU. Both models start from the same trained encoder, complete 960 additional updates and use the same checkpoint rule with all weights trainable. Memory adds 513 parameters, and the arms select checkpoints independently. Detailed results and training lineage are in the supplement.

## 5 Results

### 5.1 Genomic Assignment across Pipelines

TreeLoom attains $97.08 \pm 0.57\%$ modern span accuracy, compared with $94.45 \pm 1.15\%$ for the evaluated ARGMix release and $93.78 \pm 0.84\%$ for the derived AncestralPaths endpoint (Table 1). The difference from this ARGMix condition is 2.63 percentage points of correctly assigned sequence. TreeLoom also has the highest boundary F1 among these three fitted pipelines.

### 5.2 Modern and Ancient Query Cohorts

TreeLoom has the highest measured span accuracy in every listed cohort (Table 2). Accuracy is higher in the simulated ancient cohorts than in modern queries. These cohorts differ in sampling time, ancestry mixture and tract structure, so this pattern does not by itself indicate better handling of empirical ancient DNA. The small BAA and Yam groups also provide limited evidence about ancestry transitions.

### 5.3 Correctly Assigned Span and Tract Recovery

High span accuracy does not imply faithful ancestry tracts. TreeLoom's boundary F1 is 18.38% despite its 97.08% span accuracy. Missed, extra or displaced transitions can affect

| Cohort | Query haplotypes | AncestralPaths | ARGMix | TreeLoom |
|---|---|---|---|---|
| Modern | 182 | $93.78 \pm 0.84$ | $94.45 \pm 1.15$ | $\mathbf{97.08 \pm 0.57}$ |
| Bronze Age | 162 | $94.76 \pm 1.18$ | $97.71 \pm 0.75$ | $\mathbf{99.16 \pm 0.43}$ |
| BAA | 14 | $96.94 \pm 1.32$ | $98.44 \pm 1.02$ | $\mathbf{99.35 \pm 1.04}$ |
| Neo | 302 | $97.57 \pm 0.27$ | $99.30 \pm 0.26$ | $\mathbf{99.73 \pm 0.07}$ |
| Yam | 26 | $95.43 \pm 1.41$ | $99.22 \pm 0.90$ | $\mathbf{99.94 \pm 0.12}$ |
| Ancient | 504 | $96.54 \pm 0.32$ | $98.76 \pm 0.31$ | $\mathbf{99.55 \pm 0.13}$ |
| All queries | 686 | $95.81 \pm 0.45$ | $97.61 \pm 0.50$ | $\mathbf{98.89 \pm 0.23}$ |

Table 2: Cohort span accuracy in percent, mean±sample SD across the same eight development simulations. AncestralPaths uses the derived four source endpoint, and ARGMix uses the current author release. Bold marks the highest mean per row. Ancient and all queries aggregate the corresponding cohorts above them. The Bronze Age roster is not established as identical to the ARGMix paper's Post-Neolithic panel. Cohorts differ in sampling age, ancestry mixture and tract structure, so high simulated ancient accuracy does not establish robustness to empirical ancient DNA.

many tracts while changing the ancestry of relatively little sequence. The positional boundary score is also permissive about the source labels on either side of a transition.

The matched memory result reinforces that distinction. TreeLoom-NoMem attains $97.00 \pm 0.50\%$ accuracy and $19.05 \pm 1.43\%$ boundary F1. Adding sample memory improves accuracy by only 0.0778 percentage points, or $+0.08$ after rounding, while boundary F1 falls by $0.67$ points to $18.38 \pm 1.31\%$. The selected TreeLoom model leads span accuracy under this training procedure, but the memory comparison does not show improved tract recovery. Table 5 in the supplement gives the matched result.

## 6 Discussion

TreeLoom combines information within complete inferred genealogies with continuity along sampled chromosomes. The development results support this fitted pipeline under the stated demographic formulation.

Ancestry boundaries remain a limitation because transitions are restricted to inferred tree intervals. More flexible locations and objectives for tract recovery require evaluation. Empirical ancient genomes also introduce uncertainty in phasing, genotypes, reference sources and genealogical reconstruction. The supplement reports an earlier Danish application and genotype affinity counterevidence. These descriptive results do not validate local ancestry or calibrate predictions as mixture fractions.

**Scientific TODO.** Complete independent European tests, additional optimizer seeds and the matched American comparison. Resolve the comparator models, inputs and evaluators, plus the American demographic and ploidy assumptions, before claiming publication reproduction.

**Code availability.** Source code, experiment configurations and complete results are available at https://github.com/alexeyshmelev/ARG-GNN.

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# Supplementary Material

## A Simulation Assumptions and Parameters

The European simulations adopt the released AncestralPaths demographic formulation associated with Irving-Pease et al. (2024). The numerical assumptions below define this study condition. They are not demographic estimates inferred from the simulated observations. A census at 350 generations defines four source ancestry supervision before the older mergers of source populations. This truth census is distinct from the model's observed tree cut at the same nominal time.

The panel has 343 query and 133 diploid individuals from pure reference populations. Queries comprise 91 modern, 81 Bronze Age, seven BAA, 151 Neo and 13 Yam diploids. Pure references comprise 51 WHG, 43 EHG, 25 Ana and 14 CHG diploids. Modern samples have age zero, and ancient samples retain the released age roster in generations. The resulting 952 input haplotypes include 686 queries.

| Time (generations) | Descendant | Ancestral proportions |
|---|---|---|
| 166 | Present/Bronze | 0.5 Neo, 0.5 Yam |
| 169 | BAA | 0.25 CHG, 0.75 Ana |
| 177 | Yam | 0.5 CHG, 0.5 EHG |
| 259 | Neo | 0.25 WHG, 0.75 Ana |
| 600 | WHG, EHG | Shared NE ancestor |
| 800 | CHG, Ana | Shared WA ancestor |
| 1,500 | NE, WA | Common trunk |

Table 3: European demographic events adopted from the released formulation associated with AncestralPaths (Irving-Pease et al. 2024). Times are measured backward from the present. Proportions specify ancestral contributions at the corresponding event.

Effective population sizes are 50,000 for Present/Bronze and Neo, 20,000 for Yam and BAA, 3,000 for EHG and CHG, 2,000 for WHG, and 5,000 for Ana. The older NE, WA and trunk populations each have size 15,000. Growth and migration between events are zero. The simulation uses the Hudson coalescent, discrete genomic positions, a uniform recombination rate of $10^{-8}$ per base per generation, and JC69 mutations at $1.25 \times 10^{-8}$ per base per generation (Baumdicker et al. 2022). The 2 Mb sequence length and 32/8 training/validation split define the present development panel.

Relate uses the observed biallelic variants and sampling ages (Speidel et al. 2019). Its initial effective haploid size is 30,000, mutation rate is $1.25 \times 10^{-8}$ and generation time is 28 years. The European inference includes five iterations of population size estimation followed by estimation of branch lengths. All applicable completed model adapters consume this same final reconstruction. Mutation agreement features use observed alleles on the inferred trees.

The demographic events and population sizes follow the released AncestralPaths simulator and the Supplementary Information of Irving-Pease et al. (2024), particularly Figure S7 and pages 25, 27–28 and 30–32. Pulse times in the release are one generation older than the oldest sample in each corresponding admixed cohort. This gives 169 generations for BAA, whereas the paper reports 170. We use the released value. The basal merger is fixed at 1,500 generations as in the code. The source supplement also gives 15,000 years for that event, an unresolved difference in units. These are adopted demographic assumptions.

The complete released roster supplies the sampling ages of all 476 diploid individuals. Modern ages are fixed at zero. Ancient ages retain their individual values rather than cohort means. Table 4 summarizes their ranges. The released example uses 2 Mb sequences, while the AncestralPaths paper describes 200 Mb sequences, three training simulations and five evaluation simulations. Our 32/8 split is a separate development design.

| Cohort | Diploid individuals | Age range, generations |
|---|---|---|
| Modern | 91 | 0 |
| Bronze Age | 81 | 100–165 |
| BAA | 7 | 136–168 |
| Neo | 151 | 139–258 |
| Yam | 13 | 153–176 |
| WHG | 51 | 210–529 |
| EHG | 43 | 155–402 |
| Ana | 25 | 217–360 |
| CHG | 14 | 184–473 |

Table 4: Complete simulation panel summarized from the released AncestralPaths age roster. The first five cohorts are queries. The last four are pure source references. Ages are retained for each individual, and both phased haplotypes share that individual's sampling age.

Our simulations introduce no additional genotype error. The map is synthetic and uniform at 1 cM/Mb. The sequence occupies $[0, 2{,}000{,}000)$ in coordinates starting at zero. Variant positions are shifted by one for VCF input and shifted back after inference. Scoring uses fixed 100 bp bins and clips them to inferred coverage, leaving uncovered terminal sequence unscored. The chromosome label used for interchange does not identify an empirical chromosome or genome assembly. Complete individual ages and the fixed ancestry, mutation and inference seeds are included with the released experiment configurations.

The April ARGMix benchmark instead uses 1,611 individuals, a chromosome 22 recombination map, genotype error at rate 0.001, admixed informative references and demographic perturbations (Shanks et al. 2026). It excludes BAA. Our unperturbed 476-person condition therefore supports a comparison of fitted pipelines on common data, rather than replication of that benchmark.

**Scientific TODO.** Resolve the April ARGMix empirical map identity and coordinate domain, complete individual age roster, genotype error mechanism, reference eligibility and demographic perturbation draws before reproducing its simulation experiment. For the American benchmark, provide a primary source demographic table and resolve the haplotype interpretation and focal coordinate assumptions.

# B Model and Training Parameters

## B.1 Structural Residual Definitions

The residuals use the descendant sets and representative nodes defined in Section 3.3.

A node's gained flag $g_v$ is one when its proper clade is absent from the preceding tree. Its lost flag $\ell_v$ is one when the clade is absent from the following tree. Flags at the missing sequence neighbors are zero. The flags condition a local residual,

$$\Delta_v^{\mathrm{change}} = O[g_v(Uh_v \odot Vh_v) - \ell_v(Uh_v \odot Wh_v)]\,. \quad (10)$$

The linear projections pass through a lower dimensional space. A second residual acts on clades present in at least two consecutive trees,

$$\Delta_v^{\mathrm{local}} = \chi_v O_l[(U_l h_v) \odot (V_l h_v) + W_l h_v]\,, \quad (11)$$

where $\chi_v$ marks clades present in at least two consecutive trees. Both residuals transform the current node representation. Neighboring trees determine the structural flags, while the features being transformed come from the current tree.

## B.2 Architecture and Optimization

The genetic site encoder has two hidden layers of width 32. For a called allele it receives donor log frequency features, corresponding availability indicators, panel frequency, site entropy and a call indicator. Eight donor groups combine the four reference ancestries with ages below or at least 350 generations. Allele frequencies use additive pseudocount 0.5. If a represented reference class has no called alleles at a site in an age group, all represented classes in that age group are masked for that site. Mean and maximum pooling are combined with call count, missing pool status and interval span.

The inferred tree cut for sample memory is fixed at 350 generations. Samples older than the cut retain their own sampled node. Tree states have width 32 and local residual projections have rank eight. The memory projection also has rank eight and adds 513 parameters. Its length scale starts at 5 kb and is constrained to 100 bp–100 kb. Residual output projections start at zero. The genomic GRU has one layer and width 32 in each direction. The ancestry classifier receives 96 features, maps through width 32 and produces four scores. TreeLoom has 32,325 parameters. TreeLoom-NoMem has 31,812.

Training minimizes physical span cross entropy over all 686 queries,

$$\mathcal{L} = -\frac{\sum_{tqc} M_{tqc} \log p_{tqc}}{\sum_{tqc} M_{tqc}}, \qquad M_{tqc} = \sum_{i:r(i)=t} m_{iqc}. \quad (12)$$

Here $r(i)$ maps a scored bin to its inferred tree interval. Summing target masses within an interval preserves the original objective because the model's prediction is constant there. A complete simulated chromosome is one optimization example.

AdamW uses learning rate $10^{-3}$, weight decay $10^{-4}$, betas $(0.9, 0.999)$ and epsilon $10^{-8}$. Gradient norm is clipped at one. The reported seed is 101. The shared initializer is trained over three successive stages of 30 passes on the 32 training simulations, totaling 2,880 updates as tree processing and local conditioning are introduced. These successive stages are a training lineage, not matched component ablations. Optimizer moments reset at each stage.

For the final comparison, both TreeLoom and TreeLoom-NoMem start from the same model trained for 2,880 updates and complete 30 further passes, or 960 new updates. All inherited weights remain trainable. Checkpoints every five passes are ranked by equal simulation modern validation accuracy, then lower NLL, then earlier pass. TreeLoom selects added pass 30, or 3,840 cumulative updates. TreeLoom-NoMem selects added pass 20, or 3,520 cumulative updates, while its run also completes 3,840 updates. The alternate memory rules below share the same initializer, capacity and completed budget as TreeLoom.

# C Comparator Conditions and Unresolved Recipe Differences

**AncestralPaths.** The executed archive contains a flattening operation on a $5 \times 9$ representation, a dense layer of width 512 with ReLU, dropout 0.2 and six softmax outputs. This differs from the convolutional classifier described in the paper (Irving-Pease et al. 2024). The fit completes five epochs with Adam learning rate 0.001, batch size 32 and shuffling. A deterministic seed of 101 is supplied by the adaptation. The final epoch is used without validation checkpoint selection. Modern deterministic compatibility does not establish numerical equivalence to the original legacy execution.

The bounded sampler accepts 5,190,874 of 5,200,000 requested rows with replacement. All 9,126 missing rows are BAA path six examples, amounting to 2.2815% of requested BAA rows and 0.1755% of all requested rows. There are 53,214 replacement duplicates. Other cohort quotas are met. The sampler stops after no progress or a bound of 128 passes, so its realized shortfall is a material adaptation whose accuracy effect has not been measured. Genomic training diversity is $32 \times 2$ Mb here, whereas the paper describes $3 \times 200$ Mb. Completing five epochs does not establish convergence, and training loss is still decreasing at epoch five.

The four source probability collapse in the main text is separate from six path prediction, a retrained four path task, or evaluation restricted to non BAA routes. It must not be compared directly with a published native six path accuracy. Strict NLL is infinite in the present derived endpoint because a true source receives zero probability over 800 query base pairs. No probability floor is applied.

**ARGMix.** The evaluated author release uses 15 references without inferred ancestor nodes. Its transformer, bidirectional GRU and native training procedure are retained (Shanks et al. 2026). It uses chunks of 1,024 rows with stride 512, yielding 21,952 training and 5,488 validation chunks. All 686 queries are available, with no local subgraph cap. Training drops the nearest reference, validation drops none, label smoothing is 0.1 and loss weighting is absent. Source seed 50 is retained. The model uses supervision at the first SNP of an interval and averages overlapping softmax predictions arithmetically.

Its native selection objective is the smoothed, masked and unweighted mean validation batch loss. The selected epoch is nine of 11 completed epochs, with patience two and a cap of 20 epochs. The external span scorer does not replace this inner selection rule. The model with 25 references and three ancestor inputs described in the preprint is not recovered by the current release, whose supported loaders and extraction reject ancestor inputs. A current release run with 25 references would still have zero ancestors and would be a separate sensitivity condition. The original model, exact reference eligibility, checkpoint and publication evaluator remain unresolved. Differences in the observed panel, map, error process, training diversity and prediction support also remain material.

**American comparators.** A separate American benchmark uses a 30 Mb simulation with 600 reference and 1,000 query haplotypes from EUR, EAS and AFR sources. Native FLARE calls score $98.0100\%$ accuracy and $97.7151\%$ macro F1 on 10,000 predetermined focal records from one validation simulation. This metric counts focal records rather than physical span and has no between simulation SD. It therefore cannot be compared numerically with the European span endpoint or establish a comparative ranking on its own.

The completed FLARE condition uses author release v0.6.0 with native defaults: minimum minor allele count 50, minimum minor allele frequency 0.005, expectation maximization enabled, ten generations, ancestry proportion updates disabled, array mode disabled and seed $-99999$. Native filtering retains 27,558 of 228,477 observed markers. The evaluator uses original hard ancestry calls at the nearest native marker, with midpoint boundaries, on all 10,000 fixed focal records. It does not score a filtered subset chosen after the outcome. All prediction identities and coordinates are checked before private truth is opened.

ARGformer's completed American ancestry pipeline remains unavailable. Its published PCA/clustering classifier and nearest reference retrieval are distinct downstream routes (Bonet et al. 2026). Exact clustering, class mapping and American retrieval settings are not recovered. The prepared route retains the author release's learned normalized embedding and uses nearest reference retrieval as a disclosed reconstruction. The paper states contrastive temperature 0.05 and inverse frequency anchor weighting, whereas the release configuration uses 0.30 and its stock route can use uniform weights. These differences remain unresolved. Pretraining alone does not supply ancestry accuracy.

**Scientific TODO.** Complete independent European tests, additional optimizer seeds and larger panels. For publication reproduction, recover the original ARGMix model, reference roster, map, genotype error settings and evaluator, plus the stated AncestralPaths convolutional model and evaluator. Assess the AncestralPaths sampling deficit and ARGMix reference restriction. Complete the matched American comparison using common observations, reference and query identities, focal records and truth support, resolving its demographic and ploidy reconstruction and the published ARGformer downstream evaluator and objective differences.

| Matched model | Accuracy ↑ | Boundary F1 ↑ |
|---|---|---|
| TreeLoom-NoMem | $97.00 \pm 0.50$ | $\mathbf{19.05 \pm 1.43}$ |
| TreeLoom | $\mathbf{97.08 \pm 0.57}$ | $18.38 \pm 1.31$ |

Table 5: Effect of added sample memory after the same initialization and additional training budget. Values are mean±simulation SD in percent on the same modern endpoint. TreeLoom-NoMem retains the final genomic GRU and both complete tree traversals. Checkpoints are selected separately within each arm. Bold identifies the higher mean per metric.

| Conditioning context | Accuracy ↑ | Boundary F1 ↑ |
|---|---|---|
| Own node | $96.82 \pm 0.61$ | $18.06 \pm 1.78$ |
| Other trees | $\mathbf{96.83 \pm 0.59}$ | $\mathbf{18.16 \pm 1.38}$ |

Table 6: Matched test of context from other trees at an earlier shared initialization, with 960 added updates per arm. Values are mean±simulation SD in percent on modern queries. Both arms select pass 30. This is a conceptual context comparison, not a comparison of final training budgets with TreeLoom.

# D Additional Scientific Results

## D.1 Matched Sample Memory Comparison

TreeLoom-NoMem removes sample memory and retains both complete tree traversals, structural residuals and the final genomic GRU. Both models start from the same trained encoder and complete 960 additional updates with all weights trainable. Checkpoints are selected independently within each arm. The memory arm has 513 additional parameters, so the comparison evaluates the complete memory addition under this procedure rather than isolating history from parameter count.

TreeLoom has slightly higher accuracy and lower boundary F1. The exact paired accuracy difference is 0.0778 percentage points, rounded to 0.08 in the main text, and the mean boundary F1 difference is $-0.67$ points. These two measures address different aspects of an ancestry reconstruction. Higher accuracy for genomic span is insufficient evidence of improved tract recovery.

## D.2 Limited Matched Tests of Additional Context

An additional comparison replaces the current node context in Equation 11 with a span weighted average from other trees containing the same proper clade. Let $E(i)$ be the maximal consecutive run of trees containing that clade, and let $i$ be one member,

$$\bar{h}_{-i} = \frac{\sum_{j \in E(i)} w_j h_j - w_i h_i}{\sum_{j \in E(i)} w_j - w_i}. \tag{13}$$

Only runs with at least two trees contribute. Replacing $V_l h_i$ with $V_l \bar{h}_{-i}$ leaves the parameter count, nodes receiving the transformation, initialization and added training budget unchanged. This comparison tests the value of information from other trees beyond an extra local transformation.

| Memory rule | Accuracy ↑ | Boundary F1 ↑ |
|---|---|---|
| Group transport | $97.05 \pm 0.58$ | $18.85 \pm 1.05$ |
| Intersection transport | $97.06 \pm 0.56$ | $18.93 \pm 1.42$ |
| TreeLoom (sample memory) | $\mathbf{97.08 \pm 0.57}$ | $18.38 \pm 1.31$ |
| TreeLoom-NoMem | $97.00 \pm 0.50$ | $\mathbf{19.05 \pm 1.43}$ |

Table 7: Matched memory rules, with the same initialization and 960 added updates per arm. Values are mean±simulation SD in percent. The first three arms each have 32,325 parameters and select added pass 30. The control has 31,812 parameters and selects pass 20. Bold identifies the largest mean per metric.

The paired accuracy difference is $0.0096 \pm 0.1527$ percentage points. The paired boundary F1 difference is $0.1001 \pm 0.9632$ points. The accuracy gain falls below the registered practical threshold of 0.10 points.

We also compare sample memory with two alternatives of the same capacity. Group transport pools history through the intermediate cut group when group membership changes. Intersection transport retains the intersections of consecutive groups as the units of transport. Both use the same context from the first traversal, distance weighting, second traversal and final genomic readout as sample memory.

Intersection transport changes accuracy by $+0.0137 \pm 0.0243$ points relative to group transport and by $-0.0138 \pm 0.0384$ points relative to sample memory. It fails both registered practical criteria against every control: at least 0.10 accuracy points gained without mean boundary F1 loss, or at least one boundary F1 point gained with at most 0.05 accuracy points lost. These are development utility thresholds, not statistical significance tests.

### D.3 Descriptive Application to Ancient Danish Genomes

An earlier TreeLoom condition using context from other trees containing the same clade was applied without further fitting to ten ancient Danish query individuals and eight proxy reference individuals from the imputed genomes analyzed by Irving-Pease et al. (2024). This is the context comparison in Table 6, rather than the final model with sample memory. Five earlier individuals date to 5.44–5.67 ka cal BP and five later individuals to 3.97–4.26 ka cal BP. Three fixed 10 Mb windows on chromosomes 20–22 contain 1,158 inferred trees and 8,587 qualified site records.

Assignments weighted equally across people integrated over physical span are 14.27 percentage points higher for EHG and 5.52 points higher for CHG in the later group. EHG has a positive contrast on each chromosome. CHG depends on chromosome 21 and changes to $-0.68$ points when it is omitted. Separate genotype affinity contrasts from the same imputation resource do not support the registered directions. There is no local ancestry truth or calibration of the model's outputs as mixture fractions. These descriptive predictions therefore illustrate application to empirical genealogies and do not validate the historical interpretation.

### D.4 County Forecasting with the Adapted Core

We adapt TreeLoom's two shared GNN passes, memory and recurrent core to forecast UCI Hungarian Chickenpox Cases[1]. Six 40 epoch fits, with three initialization seeds per arm, total 4,800 updates. Forecasts use 52 observed weeks and validation MAE selection. TEST covers 20 counties and 79 weeks.

| Condition | MAE ↓ | RMSE ↓ |
|---|---|---|
| TreeLoom core | $13.908 \pm 0.243$ | $\mathbf{24.269 \pm 0.094}$ |
| NoMem core | $\mathbf{13.648 \pm 0.051}$ | $24.408 \pm 0.106$ |
| Last value | 17.497 | 31.412 |
| Seasonal lag 52 | 20.022 | 33.840 |
| Ridge ($\alpha = 1000$) | 18.959 | 27.298 |

Table 8: Errors in cases, mean±sample SD across seeds. Bold marks the lowest mean.

Memory fails the prespecified 2% MAE and paired uncertainty criteria despite lower RMSE (Table 8). Paired Full minus NoMem MAE is $+0.260$ cases, with a conditional 95% interval $[0.105, 0.427]$ from 10,000 circular resamples of 13 weeks on this fixed graph and series. Shuffling the first 51 history weeks while fixing the latest week raises both MAEs, supporting reliance on order.

[1] https://doi.org/10.24432/C5103B.